

Criminal Involvement and Health Insurance Coverage Among Serious Youth Offenders Evidence from The Pathways to Desistance Study

Author: Stefano Josue Zurita Perez

Student Id:245821990

Supervisor: Dr. Antonella Mancino

Course Coordinator: Dr. Maria Gallego

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Abstract

This paper examines the relationship between early criminal involvement and health insurance coverage among serious juvenile offenders during the transition to adulthood. Using data from the Pathways to Desistance study a longitudinal survey of 1,354 serious juvenile offenders from Phoenix and Philadelphia (US) followed into early adulthood the analysis estimates pooled OLS and individual fixed-effects models across four annual survey waves. The empirical strategy progressively adds controls to assess whether changes in criminal behavior are associated with changes in insurance status within individuals over time.

In the simplest pooled specification, any criminal involvement is associated with a 6.0 percentage point lower probability of having health insurance. This coefficient falls to -0.025 once basic controls and wave fixed effects are added, declines further to -0.008 with a full set of controls, and becomes small and statistically insignificant (0.011) in the preferred fixed-effects model. The same pattern holds when criminal involvement is measured by number of offenses. These results suggest that the negative cross-sectional association between crime and insurance is driven by persistent differences across individuals rather than within-individual changes in criminal behavior, and that labor market attachment is a more relevant correlate of insurance outcomes than criminal involvement itself.

1 Introduction

Criminal involvement at a young age can generate lasting economic and social disadvantages that extend beyond legal consequences. In addition to affecting future employment opportunities, early contact with the criminal justice system may weaken access to institutions that support long-term well-being, including health insurance. This issue is economically relevant because health insurance is a primary determinant of medical care access. In the United States, coverage is closely linked to labor market attachment, educational enrollment, and eligibility for public programs (Markowitz et al., 1991). For justice-involved youth, disruptions in these channels may make it harder to get and keep insurance coverage.

This paper investigates the relationship between early criminal involvement and health insurance coverage among youth. More specifically, it addresses the following research question: How is early criminal involvement associated with health insurance coverage among youth? The main hypothesis is that higher levels of criminal involvement are associated with a lower probability of being insured. This expectation follows from the idea that criminal activity can weaken attachment to legal employment, reduce access to employer-sponsored insurance, and make successful integration into formal social institutions more difficult.

The analysis uses data from the Pathways to Desistance study, a longitudinal survey of 1,354 serious juvenile offenders recruited through courts in Phoenix, Arizona, and Philadelphia, Pennsylvania, in the United States, and followed for up to seven years. The empirical analysis focuses on four annual survey waves during which respondents are, on average, 21 years old. The empirical strategy proceeds sequentially. It begins with pooled OLS models and then progressively adds controls before estimating individual fixed effects models. The second specification adds basic demographic and health controls, legal employment, age, age squared, and wave fixed effects. The third specification further incorporates a richer set of variables from the Pathways to Desistance study, including measures of future orientation, perceived opportunity and success, the chance of staying out of trouble, low perceived chances of success, school aspirations, baseline cognitive ability, and parental education. The final specification introduces individual fixed effects to exploit within-person variation over time and account for all time-invariant individual characteristics.

In the baseline OLS specification, any criminal involvement is associated with a statistically significant 6 percentage point reduction in the probability of being insured. This estimate shrinks and loses statistical significance as controls are added, and the coefficient becomes close to zero in the fixed effects model. This pattern holds for both the extensive margin of criminal involvement, measured by whether any crime occurred, and the intensive margin, measured by the number of offenses, with coefficients declining in magnitude across specifications and becoming statistically insignificant in the richer models. These results suggest that the negative cross-sectional association is largely driven by persistent differences across individuals rather than within-person changes in criminal involvement over time.

The rest of the paper is organized as follows. Section 2 reviews the literature. Section 3 presents the economic rationale and hypotheses. Section 4 explains the research design and methodology. Section 5 describes the data. Section 6 presents the results. Section 7 concludes.

2 Literature Review

This section reviews the existing literature related to health insurance coverage and criminal behavior. The review is organized into three main strands. First, it examines studies that analyze the socioeconomic determinants of health insurance coverage, focusing on factors such as employment, income, and demographic characteristics. Second, it discusses research that explores the relationship between criminal involvement and health outcomes or access to healthcare. Third, it reviews studies that investigate how health insurance or healthcare access may influence criminal behavior. Together, these strands help situate the research question of this study and highlight the gap addressed in this paper, namely whether criminal involvement is associated with changes in health insurance coverage among serious young offenders.

2.1 Determinants of healthcare access and coverage

Health insurance coverage is influenced by socioeconomic and institutional factors. In the economic literature, demand for health and healthcare is commonly analyzed using the framework developed by Michael Grossman in the health capital model (1972), which conceptualizes health as a form of capital that individuals invest in through medical care and other behaviors. Within this framework, access to healthcare and insurance coverage depends on individual resources, labor market participation, and socioeconomic conditions (Grossman,

1972). Empirical studies have consistently shown that employment stability, income, and education play central roles in determining whether individuals obtain health insurance coverage.

Building on this framework, Markowitz et al. (1991) analyze the determinants of health insurance coverage among young adults aged 18–24 using data from the National Medical Care Utilization and Expenditure Survey in the United States. Using a logistic regression model, the authors estimate how different socioeconomic characteristics affect the probability that an individual has health insurance coverage. Their results show that employment status is the strongest predictor of coverage. Relative to unemployed individuals, young adults working permanently and full-time are more than three times as likely to have insurance (odds ratio 3.33). Income and education also play important roles. Individuals earning more than \$25,000 annually are approximately twice as likely to be insured compared to those earning less than \$3,000, while higher educational attainment is associated with consistently higher coverage rates. The study also finds that individuals belonging to minority racial or ethnic groups have lower probabilities of being insured, reflecting disparities in access to stable employment and employer-sponsored insurance. Because the analysis focuses on individuals aged 18–24, a demographic group similar to the population examined in this study, these findings highlight how transitions into the labor market and differences in socioeconomic opportunities can strongly influence insurance coverage.

Similar socioeconomic mechanisms are observed in other institutional contexts. Duku (2018) examines the determinants of health insurance enrollment among working-age adults using household survey data from 4,214 individuals in two regions of Ghana. The study estimates logistic regression models using cross-sectional survey data collected in the early 2010s, during a period when Ghana's National Health Insurance Scheme (NHIS) was expanding coverage through a publicly administered system financed by taxes and premiums. The results indicate that socioeconomic characteristics strongly influence enrollment in the public insurance program. Individuals in the richest wealth quintile are substantially more likely to be insured than those in the poorest quintile, with adjusted odds ratios ranging from 1.73 to 2.04 depending on the region and whether individuals live in rural or urban areas. In the context of a logistic regression model, these odds ratios indicate that individuals in higher wealth groups have roughly 70 to 100 percent higher odds of being enrolled in the insurance program relative to the poorest group. Education also plays an important role: individuals with secondary education or

higher are between 1.49 and 2.07 times more likely to be insured than those with no formal education. These findings suggest that even within a publicly administered insurance system designed to expand coverage, enrollment remains strongly associated with income, education, and geographic access to healthcare services.

Evidence from other developing country settings also emphasizes the importance of socioeconomic conditions and labor market status for insurance coverage. Chauluka et al. (2022) examine the determinants of health insurance coverage among women aged 15–49 using cross-sectional data from the 2015–2016 Malawi Demographic and Health Survey. Estimating binary logistic regression models, the authors show that education, employment type, and household wealth are key predictors of insurance coverage. Women with secondary or higher education are substantially more likely to have health insurance than women with primary or no formal education. For example, 18.3 percent of women with higher education report having health insurance, compared to only 3.0 percent among women with lower levels of education. Employment conditions also matter. Women working in formal sector jobs are approximately 2.39 times more likely to be insured than those working in informal employment, reflecting the role of employer-based coverage and formal labor market participation in facilitating access to insurance. Household wealth exhibits the strongest gradient: women in the richest households are nearly sixteen times more likely to be insured than women in the poorest households. These results reinforce the broader pattern observed across countries that access to health insurance is closely linked to education, stable employment, and economic resources.

Taken together, these studies show that access to health insurance is systematically shaped by labor market participation, socioeconomic status, and institutional factors within healthcare systems. Individuals with stronger attachment to formal employment and higher levels of human capital are consistently more likely to obtain insurance coverage. However, this literature focuses primarily on socioeconomic determinants of insurance coverage and does not explicitly examine how criminal involvement may affect individuals' ability to obtain or maintain health insurance. This gap is particularly relevant for populations with prior involvement in crime, whose employment opportunities and economic stability may be disrupted during the transition to adulthood.

2.2 Effects of crime on health, health behavior, and access to healthcare

Existing research examines how criminal involvement affects health outcomes and related behaviors, highlighting links between contact with the criminal justice system, health, labor market attachment, and access to healthcare and health insurance.

Using a dynamic structural model, Gilleskie et al. (2025) study how criminal records affect health outcomes through both direct and indirect life-cycle channels. The authors jointly estimate dynamic equations for criminal justice interactions (charges, convictions, and incarceration), employment, welfare participation, education or training, and both general and mental health. The analysis uses longitudinal data on 4,033 disadvantaged women in the United States followed for up to 13 years, generating a panel of 18,672 total observations. Their results show that criminal records produce persistent negative effects on health, operating largely through disruptions to employment, education, and participation in welfare programs.

The magnitude of these effects varies depending on the type of criminal justice interaction. A recent criminal charge without conviction increases the probability of experiencing depression by approximately four percentage points. When a charge results in a conviction, the probability of depression rises by roughly seven percentage points and general health declines by about 0.5 standard deviations. Drug-related convictions produce even larger impacts, lowering general health by nearly one full standard deviation. The authors also find that past charges and convictions continue to affect health outcomes several years after the initial event, although the effects gradually decline over time. These findings provide evidence that criminal records can generate long-lasting health disadvantages, primarily by weakening individuals' attachment to employment and other socioeconomic institutions.

Criminal justice interventions can also affect mental health trajectories through institutional environments within correctional systems. Bhuller et al. (2025) provide causal evidence on the effects of incarceration on mental health using administrative data from Norway. Exploiting quasi-random variation in the timing of court decisions within an event-study framework, the authors compare individuals sentenced to prison with similar defendants whose sentencing occurred later. They find that incarceration reduces the probability of a mental health-related healthcare visit by approximately 19.5 percent during months 13–60 after sentencing, relative to pre-sentence levels. Five years after sentencing, the probability of any healthcare visit falls by

about 10 percent. These effects are driven primarily by reductions in mental health–related visits, with little change in physical health visits. The improvements are strongest in open prisons and in facilities offering education, work, or rehabilitation programs, suggesting that institutional conditions within correctional systems can significantly influence long-term mental health outcomes.

Beyond direct health effects, exposure to crime has also been shown to alter health-related behaviors that influence long-run health outcomes. Janke et al. (2016) use panel data covering nearly one million individuals across more than 320 local authority areas in England to estimate the causal impact of violent crime on physical activity. Exploiting both within-area variation over time and exogenous shocks generated by highly publicized violent events, they find that increases in local violent crime lead to measurable reductions in physical activity. In particular, the authors estimate that increases in violent crime reduce average walking time by approximately 30 minutes over a four-week period. These results suggest that exposure to crime can deteriorate health behaviors even in the absence of direct victimization, providing an additional pathway through which criminal environments may undermine long-run health outcomes.

Crime can also generate broader losses in individual well-being. Using survey data from more than 12,000 individuals in the United States, Cohen (2008) examines how criminal victimization affects life satisfaction. The study finds that individuals who experience a household burglary report life satisfaction scores that are 0.173 points lower than those of non-victims on a standardized life satisfaction scale. This decline is similar in magnitude to the reduction associated with moving from excellent to good health status. By contrast, changes in overall crime rates have relatively small effects on well-being: a 10 percent reduction in violent crime increases life satisfaction by only about 0.002 points. These results indicate that direct victimization has substantial negative effects on individual well-being and quality of life, suggesting that criminal exposure can affect individuals' broader welfare beyond direct economic losses.

Finally, criminal records have also been shown to generate persistent labor market disadvantages that may indirectly influence access to health insurance. Agan et al. (2025) examine whether policies that seal non-conviction criminal records improve employment outcomes using linked

administrative criminal court records and IRS tax data in the United States. The analysis focuses on jurisdictions that implemented automated record-sealing policies, allowing the authors to track employment outcomes before and after records are sealed. The results show that individuals experience substantial and persistent reductions in formal employment following criminal charges. Importantly, sealing non-conviction records does not significantly improve employment outcomes, even several years after the original charge. Non-conviction records are typically sealed five to six years after the event, yet the authors find no detectable employment improvement even for relatively recent charges approximately 18 months old. These findings suggest that criminal records create durable labor market scars that are difficult to reverse, highlighting how early criminal involvement can weaken long-term attachment to formal employment.

Taken together, this literature identifies several mechanisms through which early criminal involvement may affect access to health insurance. By influencing health outcomes, mental health trajectories, health-related behaviors, and labor market attachment, criminal trajectories may shape individuals' ability to obtain and maintain health insurance coverage during the transition to adulthood.

2.3 The effect of health and healthcare on crime

Existing research also studies the relationship between health, healthcare access, and criminal behavior. In this strand of the literature, health insurance and healthcare access are typically treated as determinants of criminal outcomes, rather than outcomes potentially influenced by criminal involvement. This creates a potential issue of reverse causality: while health conditions and healthcare access may affect criminal behavior through economic and behavioral channels, criminal involvement itself may also affect health and access to healthcare over time.

Andersen et al. (2026) provide causal evidence that severe health shocks can increase criminal behavior. Using administrative data covering the entire population of Denmark from 1995 to 2018, the authors exploit plausibly exogenous variation in the timing of cancer diagnoses in an event-study framework with individual fixed effects. The analysis shows that the probability of committing a crime increases by approximately 14 percent following a cancer diagnosis relative to a baseline annual crime rate of 0.69 percent. In absolute terms, this corresponds to an increase of roughly 0.10 percentage points in the probability of committing a crime in a given year. The

effect is initially small during the early years following diagnosis but grows over time and persists for more than a decade among individuals who survive cancer treatment. The authors identify several mechanisms behind this relationship, including income losses due to reduced labor market participation, changes in perceived punishment costs linked to lower survival expectations, and psychological distress. Importantly, the study finds that welfare programs that mitigate the financial consequences of health shocks substantially reduce the increase in criminal activity.

Complementing this evidence, Otsu & Yuen (2022) develop and estimate an equilibrium search model in which health status, labor market outcomes, and criminal behavior are jointly determined. The empirical analysis uses longitudinal data from the National Longitudinal Survey of Youth 1997 (NLSY97), which follows a nationally representative sample of individuals in the United States from adolescence into adulthood. The authors find a strong negative relationship between health and criminal activity. Improving health status from poor to good reduces the probability of committing property crime by approximately 2.3 percentage points per year. Given that the baseline probability of property crime in the sample is relatively low, this represents a substantial decline in criminal behavior. The authors report a semi-elasticity of -0.82 , meaning that a one-unit improvement in the health index reduces the probability of property crime by approximately 82 percent relative to the baseline level. The study also shows that nearly 80 percent of the relationship between health and crime operates through labor market mechanisms. Better health increases employment and labor force participation, which raises the opportunity cost of engaging in illegal activities. Within this framework, health insurance can influence criminal behavior indirectly by stabilizing health outcomes and improving labor market attachment.

Similarly, Hendrix & Stock, (2024) provide causal evidence that public health insurance coverage during childhood reduces criminal activity later in life in the United States. Exploiting variation generated by Medicaid eligibility expansions during the 1980s and 1990s, the authors estimate the long-run effects of childhood exposure to public health insurance. They find that each additional year of Medicaid eligibility during childhood reduces property crime arrests by approximately 7 percent, drug-related arrests by about 5 percent, and DUI arrests by roughly 4 percent between the ages of 19 and 24. The effects are particularly strong for individuals who become eligible during adolescence, defined as the ages immediately preceding entry into

adulthood, when individuals begin making independent decisions about employment, education, and criminal behavior. These findings highlight how access to public health insurance early in life can influence long-run criminal trajectories.

Despite extensive evidence linking health shocks, healthcare access, and insurance coverage to criminal behavior, much less is known about the reverse relationship: whether criminal trajectories influence individuals' access to health insurance over time. This question is particularly relevant for populations that experience early socioeconomic disadvantage or involvement with the criminal justice system, whose employment opportunities and health trajectories may evolve differently from those of the general population. By focusing on a sample of disadvantaged young offenders followed into early adulthood, this study examines whether criminal involvement is associated with changes in health insurance coverage during a critical stage of the life cycle.

2.4 Contribution

This paper examines how early criminal trajectories affect access to health insurance coverage during the transition to adulthood. The analysis uses data from the Pathways to Desistance project, a longitudinal study that follows serious adolescent offenders for up to seven years as they transition into early adulthood.

3 Economic Rationale

Health insurance coverage depends on eligibility rules, financial resources, and the ability to maintain stable connections to employment, education, and administrative systems. In standard health economic models, individuals demand health insurance as a means of smoothing health-related risks over the life cycle, subject to income, prices, and institutional constraints (Cutler & Zeckhauser, 2000; Grossman, 1972). Access to coverage in practice is therefore closely tied to labor market attachment, educational enrollment, and eligibility for public insurance programs (Duku, 2018).

In a dynamic setting, changes in criminal involvement may be associated with changes in insurance coverage through several channels. Criminal activity can disrupt employment and schooling, reduce earnings, and increase residential instability, all of which may affect eligibility for employer-sponsored insurance or continuity of public coverage (Markowitz et al., 1991;

Pager, 2003). Administrative barriers linked to criminal justice contact may further complicate enrollment and recertification in insurance programs.

This section outlines the economic rationale linking criminal involvement to health insurance coverage. The main prediction is that higher criminal involvement is associated with a lower probability of being insured. This expectation follows from the idea that criminal involvement may weaken the links to employment, education, and administrative systems that support access to insurance.

4 Data

This section describes the dataset used in the analysis, defines the main variables, and presents descriptive statistics and graphical evidence. It also explains how the data were cleaned and structured for the empirical analysis and highlights the main characteristics of the dataset.

4.1 Pathways to Desistance

The Pathways to Desistance (PD) study is a multi-site longitudinal study of serious juvenile offenders designed to examine how adolescents transition into early adulthood and move toward desistance from antisocial behavior. The study emerged from the planning efforts of the MacArthur Foundation Research Network on Adolescent Development and Juvenile Justice and was developed as part of a broader agenda to provide empirical evidence that could improve juvenile justice practice and decision-making, particularly in relation to competence, culpability, risk assessment, and the treatment of serious young offenders.

4.2 Data Description

The full PD sample consists of 1,354 individuals from Phoenix, Arizona, and Philadelphia, Pennsylvania, in the United States, who were first interviewed at baseline and then followed for up to seven years through 11 survey waves. Follow-up interviews were conducted every six months during the first three years and annually thereafter, creating a rich longitudinal dataset that tracks changes in offending behavior, social circumstances, and other life outcomes over time.

The analytic sample used in this paper focuses on survey waves 8 through 11, corresponding to months 48, 60, 72, and 84. These waves are used because they provide consistent and

comparable information on both health insurance coverage and self-reported criminal activity. The resulting dataset is structured as an individual-wave panel, where each observation corresponds to one individual in a given survey wave.

The PD study is particularly useful for this paper because it follows a population facing substantial barriers during the transition to adulthood, including challenges related to criminal involvement, employment, and access to social institutions. At the same time, the sample does not represent the general youth population. Instead, it focuses on a specific group of justice-involved individuals with serious offending histories, so the results should be interpreted as applying to this high-risk population rather than to all young people in the United States.

Because the same individuals are observed repeatedly across multiple waves, the data make it possible to distinguish persistent differences across individuals from changes occurring within the same person over time. This is especially relevant in a setting where both criminal involvement and health insurance coverage may evolve as individuals age and move through school, employment, and other institutions.

Attrition and Sample Composition

The estimation sample declines modestly across waves, from 1,127 observations in wave 8 (48 months) to 1,071 in wave 11 (84 months), representing approximately a 5 percent reduction. Comparing individuals observed in all four waves with those with incomplete panels reveals only limited differences in the main observable characteristics. Completers and non-completers have very similar rates of health insurance coverage (49.1 percent versus 48.9 percent) and criminal involvement (43.7 percent versus 42.7 percent), as well as similar age, gender composition, ethnicity, and study site. The main observable difference is that individuals with complete panels report slightly more weeks of legal employment on average. Overall, these comparisons suggest that attrition in the later waves does not substantially alter the observable composition of the sample, although differences in unobserved characteristics cannot be ruled out.

Self-Reported Crime: Nature of the Measure

The self-reported crime data in the Pathways to Desistance study are collected through structured interviews and reflect criminal activity reported directly by respondents, regardless of whether

those offenses resulted in arrest, charges, prosecution, or sealed records. This is an important distinction because the crime measures used here capture a broader range of criminal involvement than administrative records, including offenses that did not result in formal legal consequences. This is a useful feature of the data because it helps avoid the truncation bias inherent in arrest or conviction records. At the same time, self-reported measures may be affected by recall error or social desirability bias. Four mutually non-exclusive crime type indicators are constructed from the self-report items: violent crime, property crime, drug-related crime, and other crime. Drug-related crime refers specifically to selling drugs, while the other crime category includes behaviors such as driving under the influence, carrying a gun, and paying for sex. Because these categories are not mutually exclusive, a respondent who reports both a violent and a property offense in the same wave is coded as one in both indicators.

4.3 Variable Definitions

This subsection defines the main variables used in the analysis.

Health Insurance Coverage

Health insurance coverage is measured using the health insurance question (HC1) asked in survey waves 8 to 11. In each of these waves, respondents are asked whether they have health insurance and are provided with seven response categories. Six categories correspond to having insurance through different sources, including coverage obtained on one's own, through school, employment, a parent's plan, a spouse's plan, or public medical assistance, while the seventh category indicates that the respondent does not have health insurance.

Because insurance coverage is not recorded as a direct binary variable, responses are recoded into a dummy variable equal to one if the respondent reports any form of coverage (categories 1–6) and equal to zero if the respondent reports no coverage (category 7). Observations with missing or invalid responses, such as refusals or “don't know” answers, are excluded to ensure consistency across waves.

In addition to this binary measure, alternative variables capturing the type of insurance are constructed, including parental, private, and public coverage. These measures are used in extended specifications to explore heterogeneity in access to insurance. Self-reported overall health is not used as the primary health measure in this analysis. Instead, health status is captured using categorical indicators for excellent, good, fair, and poor health, which are included as control variables in the regression models.

Table 1 Distribution of Insurance Coverage by Type:

Insurance type	Frequen cy	Perce nt
Uninsured	2,234	50.84
Public insurance	1,208	27.49
Private insurance	647	14.72
Parental insurance	305	6.94
Total	4,394	100.00

Note: Based on survey waves 8–11. Source: Author’s calculations using the Pathways to Desistance data.

Table 1 reports the distribution of insurance coverage by type. Approximately 50.8 percent of observations correspond to individuals without insurance, while 27.5 percent rely on public coverage, 14.7 percent on private insurance, and 6.9 percent on parental insurance.

Criminal Involvement

Criminal involvement is measured using a binary indicator, `crime_exist`, which captures whether an individual reports committing at least one criminal offense during the 12 months preceding the interview. The variable equals one if the individual reports any criminal activity and equals zero otherwise.

This indicator is constructed from a set of self-reported offending questions included in the survey. These questions cover multiple categories of criminal behavior, including property crimes, drug-related offenses, violent crimes, and weapons-related offenses. The variable aggregates responses across these categories to capture overall criminal involvement within the reference period. Offenses related to homicide and forced sexual acts are excluded due to confidentiality restrictions in the public-use version of the dataset.

In addition to the binary measure, an intensive margin variable is constructed as the total number of self-reported offenses committed by an individual in each wave. This variable captures variation in the frequency of criminal activity and is used in extended specifications to examine whether the intensity of criminal behavior is associated with health insurance coverage.

4.3.1 Control Variables

To account for observable factors that may jointly influence criminal behavior and health insurance coverage, the analysis includes demographic, health, employment, and baseline controls. These variables are included to improve precision and reduce confounding and are not the primary objects of interest.

Time-varying controls (measured at each survey wave):

- **Age (age):** Continuous variable measuring the individual’s age in years at the time of each interview. Age varies across survey waves and captures life-cycle effects that may influence both criminal behavior and access to health insurance.

- **Health status:** A set of categorical indicators capturing self-reported health (excellent, good, fair, poor). These variables are included as controls to account for differences in health conditions that may affect insurance participation and economic outcomes.
- **Weeks worked legally (weeks_legal):** Continuous variable measuring the number of weeks the individual worked in legal employment during the reference period. This variable provides a more detailed measure of labor market participation.
- **Behavioral expectations:** Four variables capture forward-looking attitudes and perceptions that may jointly influence criminal behavior and insurance decisions. Future orientation (scale 1–4, higher = more future-oriented) measures the degree to which an individual considers future consequences when making decisions. Perceived opportunity for success (scale 1–5) captures aspirations for work, family, and staying out of trouble. Chance of staying out of trouble (scale 1–5, higher = more confident) reflects the respondent’s self-assessed likelihood of avoiding future criminal involvement. Low chance of success (scale 1–5, higher = more pessimistic) measures the belief that one’s chances of success in life are not very good. School aspiration (scale 1–5) records how far the respondent would like to go in school. These variables are typically unobserved in standard datasets but are available here and help reduce omitted-variable bias by controlling for stable differences in motivation and expectations.

Time-invariant demographic controls:

- **Ethnicity (ethn):** Categorical variable indicating self-reported ethnicity (White, Black, Hispanic, Other). This variable is collected in the demographic module and does not vary over time.
- **Gender (female):** Binary indicator equal to one if the individual is female and zero if male. Gender is measured at baseline and is time-invariant.
- **Study Site (phoenix):** Binary indicator equal to one if the individual was recruited in Phoenix, Arizona, and zero if recruited in Philadelphia, Pennsylvania. Study site is fixed at baseline.

Baseline socioeconomic and household characteristics (Survey 1):

- **Baseline cognitive ability (iq_base):** Continuous variable based on the Wechsler Abbreviated Scale of Intelligence (WASI) Full Scale IQ score collected at baseline. This variable captures pre-existing cognitive ability and does not vary over time.
- **Parental education at baseline (paredu_base):** Discrete variable capturing the highest level of education attained by the individual's parents, as reported in the baseline survey. Higher values correspond to higher levels of parental education and serve as a proxy for socioeconomic background prior to the outcomes analyzed in the study.

4.4 Descriptive Statistics

Table 2 presents descriptive statistics for the main variables used in the analysis. The sample consists of 4,394 individual-wave observations from survey waves 8 to 11. The table reports the mean, median, standard deviation, and range for all variables included in the empirical models.

Table 2 presents descriptive statistics for the variables used in the analysis. The data show substantial variation in both health insurance coverage and criminal behavior. Approximately 49.2 percent of observations report having health insurance, indicating that the sample is nearly evenly split between insured and uninsured individuals. Criminal involvement is also common, with 43.3 percent of observations reporting at least one offense in the previous year. The intensive margin measure shows that individuals commit, on average, 1.290 offenses per wave, with considerable dispersion.

Table 2 Descriptive Statistics

Variable	Mean	Median	S.D.	Min	Max
Health insurance coverage (insured)	0.492	0.000	0.500	0.000	1.000
Criminal involvement (crime_exist)	0.433	0.000	0.494	0.000	1.000
Number of crimes (IM_num_crimes)	1.290	0.000	2.351	0.000	20.000
Parental insurance (insured_parent)	0.069	0.000	0.254	0.000	1.000
Private insurance (insured_private)	0.147	0.000	0.354	0.000	1.000
Public insurance (insured_public)	0.275	0.000	0.447	0.000	1.000
Age (age)	21.501	21.000	1.600	18.000	26.000
Weeks worked legally (weeks_legal)	13.024	0.000	18.800	0.000	60.670

Health status: Excellent (health_excellent)	0.341	0.000	0.474	0.000	1.000
Health status: Good (health_good)	0.510	1.000	0.500	0.000	1.000
Health status: Fair (health_fair)	0.135	0.000	0.342	0.000	1.000
Health status: Poor (health_poor)	0.013	0.000	0.112	0.000	1.000
Future orientation (future_orientation)	2.691	3.000	0.547	1.000	4.000
Perceived opportunity / success (opp_success)	4.516	5.000	0.516	1.000	5.000
Chance of staying out of trouble (stay_out_trouble)	3.917	4.000	1.159	1.000	5.000
Low chance of success (low_success_chance)	2.031	2.000	0.781	1.000	5.000
School aspiration (school_aspiration)	3.344	3.000	1.000	1.000	5.000
Female (female)	0.145	0.000	0.352	0.000	1.000
Black (black)	0.393	0.000	0.489	0.000	1.000
Hispanic (hispanic)	0.343	0.000	0.475	0.000	1.000
Other race (other_race)	0.046	0.000	0.210	0.000	1.000
Phoenix (phoenix)	0.508	1.000	0.500	0.000	1.000

Note: Statistics are computed using the analytic panel consisting of survey waves 8–11 (months 48, 60, 72, and 84). The sample consists of 4,394 person-wave observations). All variables are computed for the estimation sample with complete data across all specifications. **Source:** Author’s calculations using the Research on Pathways to Desistance public-use data.

The sample reflects the demographic composition of a high-risk and disadvantaged population. Approximately 39.3 percent of observations correspond to Black respondents, 34.3 percent to Hispanic respondents, and 21.7 percent to White respondents. Only 14.5 percent of observations correspond to female respondents, which is consistent with the male-dominated composition of serious adolescent offending. These demographic characteristics reinforce that the sample is not representative of the general youth population; instead, the findings should be interpreted as applying to serious juvenile offenders in two urban U.S. settings. Socioeconomic characteristics are also consistent with a population transitioning into early adulthood. Individuals are, on average, 21.501 years old, and weeks of legal employment average 13.024 per wave, with many observations reporting zero weeks worked, suggesting limited and sporadic attachment to the formal labor market.

Health status is generally favorable. About 34.1 percent of observations report excellent health and 51.0 percent report good health, while 13.5 percent report fair health and only 1.3 percent report poor health. Among the behavioral expectation variables, perceived opportunity for success averages 4.516 on a 1–5 scale, indicating that most respondents are relatively optimistic about their prospects. By contrast, the low-success-chance variable averages 2.031, suggesting that pessimism about future success is moderate on average but varies across individuals. Future orientation averages 2.691 on a 1–4 scale. Taken together, these patterns indicate meaningful heterogeneity across individuals and over time, providing sufficient variation to examine the relationship between criminal involvement and health insurance coverage.

4.5 Graphical Analysis

Figure 1: Health Insurance Coverage and Criminal Involvement over Time

Note: Mean values by survey wave for the analytic sample (waves 8–11). **Source:** Author’s calculations using the Pathways to Desistance public-use data.

Figure 1 presents the evolution of average health insurance coverage and criminal involvement across survey waves 8 to 11, corresponding to months 48, 60, 72, and 84 of the study. The figure summarizes the mean values of both variables at each wave and complements the descriptive statistics reported in Table 2.

Two main patterns emerge. First, average health insurance coverage shows a gradual downward trend over the period considered. Mean coverage falls from 0.54 at month 48 to 0.47 at month 84, suggesting that access to insurance becomes somewhat weaker as respondents move further into early adulthood. Second, criminal involvement also declines over time, from 0.46 at month 48 to 0.39 at month 84. This pattern is consistent with the age-crime profile documented in the literature, according to which offending tends to decline as individuals age (Hirschi & Gottfredson, 1983; Lochner & Moretti, 2004).

In addition to these trends, the raw correlation between criminal involvement and health insurance coverage is negative ($\rho = -0.065$). Although this association is modest in magnitude, it suggests that respondents who report criminal activity are, on average, less likely to have health insurance. Overall, the graphical evidence indicates that both variables decline across waves and are negatively associated in the raw data. These descriptive patterns motivate the use of panel methods in the following sections to examine whether this relationship remains once the analysis accounts for individual heterogeneity.

5 Methodology, Regression Equation and Hypotheses

This section presents the empirical strategy used to examine the relationship between criminal involvement and health insurance coverage. The main objective is to assess whether changes in criminal behavior are associated with changes in insurance status within the same individual over time. To address this question, the analysis uses panel data from multiple waves of the Pathways to Desistance study and estimates a sequence of regression models that progressively add controls and fixed effects. This approach makes it possible to compare cross-sectional associations with within-individual estimates, helping to assess whether the observed relationship remains once persistent individual differences are considered.

5.1 Methodology

The strategy builds on previous literature examining the relationship between health, economic outcomes, and criminal behavior. Studies such as (Hendrix & Stock, 2024), along with related work in health economics, suggest that health shocks and economic conditions can influence criminal behavior through labor market participation, psychological distress, and financial

constraints. The empirical analysis here examines whether changes in criminal involvement are associated with changes in health insurance coverage within individuals over time.

An important concern in this setting is endogeneity. Unobserved individual characteristics such as family background, early-life disadvantage, cognitive ability, stable preferences, and risk attitudes may influence both criminal behavior and health insurance coverage. If these factors are omitted, cross-sectional estimates may confound the relationship of interest with persistent differences across individuals rather than changes in criminal behavior itself. In addition, time-varying factors such as expectations about the future and perceived opportunities may also affect both criminal involvement and insurance outcomes.

To address these concerns, the analysis proceeds in stages. It begins with pooled linear probability models, which capture the cross-sectional association between criminal involvement and insurance coverage. The specification is then extended by adding demographic characteristics, health status, weeks worked legally, behavioral expectations, and wave fixed effects. The main specification relies on individual fixed-effects models, which exploit within-individual variation over time and remove all time-invariant individual characteristics from the estimation. Comparing pooled and fixed-effects estimates helps assess whether the observed relationship reflects persistent differences across individuals or changes within individuals over time.

5.2 Empirical Specification

The main empirical specification estimates the following panel regression model:

$$Insured_{it} = \beta_0 + \beta_1 Crime_{it} + \beta_2 X_{it} + \alpha_i + \varepsilon_{it}$$

where:

- $Insured_{it}$ is a binary variable indicating whether individual i has health insurance at time t .
- $Crime_{it}$ measures criminal involvement during the period.
- X_{it} represents a vector of control variables.
- α_i captures individual-specific fixed effects.
- ε_{it} is the idiosyncratic error term.

The vector of control variables includes age, health status, weeks worked legally, and behavioral expectations such as future orientation and perceived opportunities. In the fixed-effects specification, α_i captures all time-invariant individual characteristics that may affect health insurance coverage and be correlated with criminal involvement. Because the data follow the same individuals across multiple survey waves, estimation relies on within-individual changes over time.

5.3 Expected signs of Hypothesis

The main coefficient of interest is β_1 , which measures the relationship between criminal involvement and health insurance coverage. The expected sign of this coefficient is negative. Individuals who engage in criminal behavior may be less likely to maintain health insurance coverage for several reasons. First, criminal activity is often associated with weaker attachment to the formal labor market, where many individuals obtain employer-sponsored health insurance. Second, involvement in criminal behavior may increase economic instability, reducing the ability to maintain consistent insurance coverage. Third, legal or institutional barriers may limit access to certain employment opportunities or benefits, which can indirectly affect insurance status.

Based on these mechanisms, the main hypothesis tested in this analysis is that criminal involvement is associated with a lower probability of having health insurance coverage.

The control variables are included to account for additional socioeconomic and behavioral factors that may influence insurance participation. Age captures life-cycle patterns in insurance

coverage, while weeks worked legally proxy for attachment to the formal labor market, an important determinant of access to health insurance. Health status is also included to control for differences in health conditions that may affect insurance participation.

6 Results:

The results tell a consistent story across specifications. The analysis progresses from pooled OLS to individual fixed-effects models, with regression results for the extensive margin (Table 3), intensive margin (Table 4), and crime type heterogeneity (Table 5) presented below. The key finding is that the negative association between criminal involvement and health insurance coverage disappears once individual fixed effects are introduced. The OLS association appears to be driven largely by persistent differences across individuals rather than by within-individual changes in criminal behavior over time

6.1 Criminal Involvement and Health Insurance

Table 3 reports the main results of the paper. The table presents four specifications that progressively add controls in order to assess how sensitive the estimated relationship between criminal involvement and health insurance is to observed and unobserved individual characteristics.

In column (1), the pooled OLS estimate shows that individuals who report any criminal involvement are 6.0 percentage points less likely to have health insurance. This coefficient is statistically significant and indicates a negative raw cross-sectional association between crime and insurance coverage.

Column (2) adds basic demographic controls, health status, weeks worked legally, and wave fixed effects. Once these controls are included, the crime coefficient falls to -0.025 and becomes statistically insignificant. This suggests that part of the negative association in column (1) reflects observable differences across individuals rather than criminal involvement itself. Among the controls, female respondents are about 31.9 percentage points more likely to be insured, while Black respondents are 7.1 percentage points more likely to be insured relative to the omitted White category. Age enters with a negative coefficient and age squared with a positive coefficient, indicating a non-linear relationship between age and insurance coverage over the

transition to adulthood. Weeks worked legally is positively associated with insurance, although the effect is small in magnitude.

Table 3: Criminal Involvement and Health Insurance

Variable	OLS	Basic controls + Wave FE	Complete set of controls	Complete set + FE
Criminal involvement (Any crime)	-0.060***	-0.025	-0.008	0.011
Female		0.319***	0.295***	
Phoenix		-0.046	-0.041	
Black		0.071*	0.059*	
Hispanic		-0.039	-0.021	
Other race		0.045	0.038	
Age		-0.318**	-0.321**	-0.544***
Age squared		0.007**	0.007**	0.010***
Health status: Good		-0.044*	-0.025	0.001
Health status: Fair		-0.019	0.006	0.006
Health status: Poor		-0.084	-0.066	-0.051
Weeks worked legally		0.002***	0.002**	0.001
Wave 8		0.000	0.000	0.000
Wave 9		-0.021	-0.018	0.076
Wave 10		-0.032	-0.031	0.169*
Wave 11		-0.017	-0.013	0.264*
Future orientation			0.033*	-0.003
Perceived opportunity / success			-0.007	-0.000
Chance of staying out of trouble			0.024**	0.007
Low chance of success			-0.040***	-0.025*
School aspiration			0.014	0.004
Baseline cognitive ability (IQ)			-0.002**	
Parental education at baseline			-0.029**	
Constant	0.517***	4.208***	4.434***	7.492***
Observations	4394.000	4394.000	4394.000	4394.000
R-squared	0.004	0.083	0.098	0.020

Note: The dependent variable equals one if the respondent has health insurance coverage and zero otherwise. Column (1) reports pooled OLS estimates. Column (2) adds basic controls and wave fixed effects. Column (3) adds the full set of controls, including behavioral expectations and baseline socioeconomic characteristics. Column (4) reports the fixed-effects specification. Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column (3) adds the full set of controls, including behavioral expectations and baseline socioeconomic characteristics. The crime coefficient shrinks further to -0.008 and remains statistically insignificant. This indicates that, even before introducing fixed effects, the negative association becomes very small once richer individual-level characteristics are taken into

account. Several controls remain informative in this specification. Female respondents continue to have a substantially higher probability of being insured, and Black respondents remain positively associated with insurance coverage. Age and age squared continue to show a non-linear pattern. Among the additional controls, low chance of success enters with a negative and statistically significant coefficient, while baseline cognitive ability and parental education also show negative and statistically significant associations with insurance. These signs may appear unexpected, but they likely reflect the complex composition of this high-risk sample and the fact that coverage comes from multiple sources, including parental and public insurance.

Column (4) introduces individual fixed effects, which remove all time-invariant individual characteristics from the estimation. This is the preferred specification. The coefficient on criminal involvement becomes 0.011, which is very small and statistically insignificant. The main implication is not that crime increases insurance coverage, but rather that there is no evidence of a meaningful within-individual relationship between criminal involvement and insurance status once stable individual heterogeneity is accounted for. In the fixed-effects model, age remains negative and age squared remains positive, again suggesting a non-linear pattern over time. Low chance of success also remains negatively associated with insurance coverage, although most of the other time-varying controls are statistically insignificant.

The R-squared values reflect the different sources of variation used by the OLS and fixed-effects models. In the OLS specifications, R-squared rises as additional controls are included, reaching approximately 0.10 in the richer models. This indicates that the model explains about 10 percent of the total variation in insurance status. In contrast, the fixed-effects specification yields a within R-squared of about 0.02. This lower value should not be interpreted as evidence of a weak model. By construction, fixed effects remove all stable between-individual variation and rely only on within-individual changes over time. Since insurance coverage is relatively persistent for many respondents and the panel is short, the remaining within variation is limited. For that reason, a low within R-squared is expected. The two R-squared measures are not directly comparable because they refer to different dimensions of variation. Overall, the stronger OLS relationship appears to be driven mainly by persistent differences across individuals rather than by short-run changes within the same person.

6.2 Intensive Margin of Crime

Table 4 follows the same four-column structure as Table 3, but replaces the binary measure of criminal involvement with the intensive-margin measure based on the number of offenses reported in each wave. This allows the analysis to assess whether greater intensity of offending is associated with health insurance coverage.

Table 4: Intensive Margin of Crime

Variable	OLS	Basic controls + Wave FE	Complete set of controls	Complete set + FE
Number of crimes (Intensive margin)	-0.015***	-0.008*	-0.004	-0.003
Female		0.317***	0.294***	
Phoenix		-0.044	-0.040	
Black		0.072*	0.059*	
Hispanic		-0.039	-0.021	
Other race		0.048	0.039	
Age		-0.320**	-0.322**	-0.542***
Age squared		0.007**	0.007**	0.010***
Health status: Good		-0.043*	-0.025	0.000
Health status: Fair		-0.019	0.006	0.005
Health status: Poor		-0.081	-0.064	-0.053
Weeks worked legally		0.002***	0.002**	0.001
Wave 9		-0.021	-0.019	0.074
Wave 10		-0.033	-0.031	0.164*
Wave 11		-0.018	-0.014	0.256*
Future orientation			0.033*	-0.004
Perceived opportunity / success			-0.008	0.000
Chance of staying out of trouble			0.023**	0.006
Low chance of success			-0.040***	-0.024*
School aspiration			0.014	0.005
Baseline cognitive ability (IQ)			-0.002**	
Parental education at baseline			-0.030**	
Constant	0.511***	4.224***	4.447***	7.467***
Observations	4394.000	4394.000	4394.000	4394.000
R-squared	0.005	0.084	0.099	0.020

Note: The dependent variable equals one if the respondent has health insurance coverage and zero otherwise. Column (1) reports pooled OLS estimates. Column (2) adds basic controls and wave fixed effects. Column (3) adds the full set of controls, including behavioral expectations and baseline socioeconomic characteristics. Column (4) reports the fixed-effects specification. Standard errors are clustered at the individual level. * p<0.10, ** p<0.05, *** p<0.01.

The pattern closely mirrors the results in Table 3. In the simple OLS specification, each additional offense is associated with a 1.5 percentage point reduction in the probability of being

insured, and this coefficient is statistically significant. Once basic controls are added in column (2), the coefficient falls to 0.8 percentage points in absolute value and loses most of its significance. In column (3), which includes the full set of controls, the coefficient declines further to 0.4 percentage points in absolute value and is no longer statistically significant. In the fixed-effects model in column (4), the estimate is -0.003 and remains statistically insignificant. Regardless of whether criminal involvement is measured as a binary indicator or as a count of offenses, the main conclusion is the same: the negative cross-sectional association weakens substantially as controls are added and disappears once the analysis relies on within-individual variation over time. This pattern reinforces the view that the stronger OLS relationship is largely driven by persistent differences across individuals rather than by short-run changes in offending within the same person.

The control variables show a pattern similar to Table 3. Female and Black respondents are more likely to be insured in the pooled specifications, while age and age squared suggest a non-linear life-cycle pattern. In the fixed-effects specification, low perceived chances of success remains negatively associated with insurance coverage.

6.3 Crime Type Heterogeneity: No Crime vs. Each Crime Type

Table 5 isolates within-individual differences between waves with no reported crime and waves with each specific type of crime. All four crime-type indicators are included simultaneously, so each coefficient is interpreted relative to crime-free waves, holding the other controls constant.

Property crime is the only category with a positive coefficient that is marginally statistically significant. Its estimate is 0.058, implying that, within individuals, waves with reported property crime are associated with a 5.8 percentage point higher probability of being insured relative to waves with no reported crime. Violent crime (-0.004), drug-related crime (-0.0003), and other crime (-0.029) are all statistically insignificant. Given that the positive coefficient appears only for property crime and only at the margin of significance, this result should be interpreted with caution and may reflect sampling variation rather than a true behavioral relationship.

Table 5: Heterogeneity Check Fixed Effects Model Regressed with all crime types

Variables	FE Full Model
Violent crime (vs. no crime)	-0.0038
Property crime (vs. no crime)	0.0584*
Drug-related crime (vs. no crime)	-0.0003
Other crime (vs. no crime)	-0.0288
Age	-0.5416***
Age squared	0.0098***
Health status: Good	-0.0007
Health status: Fair	0.0040
Health status: Poor	-0.0527
Weeks worked legally	0.0007
Future orientation	-0.0040
Perceived opportunity / success	0.0003
Chance of staying out of trouble	0.0066
Low chance of success	-0.0253*
School aspiration	0.0041
Wave 9	0.0761
Wave 10	0.1676**
Wave 11	0.2602**
Constant	7.4584***
Observations	4,394
Within R-squared	0.0220

Note: The dependent variable equals one if the respondent has health insurance coverage and zero otherwise. The omitted reference category is a wave with no reported criminal involvement. The model includes all four crime-type indicators simultaneously, together with age, age squared, health status, weeks worked legally, behavioral expectations, and wave fixed effects. Individual fixed effects are included in this specification. Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The control variables follow a pattern similar to the previous fixed-effects specifications. Age remains negative and age squared remains positive, again suggesting a non-linear relationship between age and insurance over the transition to adulthood. Low chance of success also remains

negatively associated with insurance coverage, while the later survey waves have positive coefficients relative to the omitted wave.

6.4 Overall Interpretation of the Results

Taken together, the results provide a consistent answer to the main research question. The negative association between criminal involvement and health insurance appears clearly in the pooled models, but it becomes much smaller and statistically insignificant once richer controls and individual fixed effects are introduced. This pattern holds when criminal involvement is measured as a binary indicator, as the number of offenses, and when it is separated by crime type. Across specifications, the results suggest that the raw negative relationship is driven mainly by persistent differences across individuals rather than by within-individual changes in criminal behavior over time.

This interpretation is also consistent with the pattern observed for the control variables. Age and age squared are jointly informative across specifications, indicating a non-linear relationship between age and insurance coverage during the transition to adulthood. Low chance of success also remains negatively associated with insurance in the fixed-effects models. At the same time, the coefficients on criminal involvement become small and unstable once the analysis focuses on within-individual variation.

The appendix results also suggest that labor market attachment is more relevant for some types of insurance outcomes than criminal involvement itself. In Appendix Table A1, criminal involvement is not statistically significant for parental insurance (0.015), private insurance (-0.005), or public insurance (0.001). By contrast, weeks worked legally is not significantly related to parental insurance (-0.000), but it is positively associated with private insurance (0.003***) and negatively associated with public insurance (-0.002***). This suggests that legal employment matters more for the type of insurance individuals hold than criminal involvement itself.

Overall, the evidence provides little support for a meaningful within-individual relationship between criminal involvement and health insurance coverage. Instead, the results point to persistent individual heterogeneity and labor market attachment as the main factors shaping insurance outcomes in this sample.

7 Summary and Conclusions:

This paper examined whether early criminal involvement is associated with health insurance coverage during the transition to adulthood among serious youth offenders. Motivated by the idea that criminal involvement may weaken attachment to employment, education, and other formal institutions that support insurance access, the analysis asked whether changes in criminal behavior are associated with changes in insurance status within individuals over time. Using data from the Pathways to Desistance study, the paper focused on four annual survey waves between months 48 and 84, when respondents were in early adulthood and insurance coverage became especially relevant. The empirical strategy compared pooled OLS estimates with richer specifications that progressively added controls and, ultimately, individual fixed effects.

The main findings are clear. In the simplest pooled specification, any criminal involvement is associated with a 6.0 percentage point lower probability of having health insurance. However, this coefficient falls to -0.025 once basic controls and wave fixed effects are included, declines further to -0.008 with the full set of controls, and becomes small and statistically insignificant in the preferred fixed-effects model (0.011). The same pattern appears when criminal involvement is measured at the intensive margin: the coefficient falls from -0.015 in the pooled specification to -0.008 with basic controls, -0.004 with the full set of controls, and -0.003 in the fixed-effects model. When crime is separated by type, property crime is the only category with a marginally significant positive coefficient, while violent, drug-related, and other crime remain statistically insignificant. Taken together, these results indicate that the negative association observed in pooled models does not survive once the analysis focuses on within individual variation over time.

The results therefore suggest that the raw negative relationship between criminal involvement and health insurance is driven mainly by persistent differences across individuals rather than by within individual changes in criminal behavior. This interpretation is consistent with the broader pattern of control variables. Age and age squared are jointly informative across specifications, indicating a non-linear relationship between age and insurance coverage during the transition to adulthood, while low chance of success remains negatively associated with insurance in the fixed-effects models. At the same time, the coefficients on criminal involvement become small and unstable once the analysis accounts for stable individual heterogeneity.

Additional appendix evidence also points more clearly to labor market attachment than to criminal involvement as a correlate of insurance outcomes. In the fixed-effects insurance-type models, criminal involvement is not statistically significant for parental insurance (0.015), private insurance (-0.005), or public insurance (0.001). By contrast, weeks worked legally is not significantly related to parental insurance, but it is positively associated with private insurance (0.003***) and negatively associated with public insurance (-0.002***). This pattern suggests that legal employment matters more for the type of insurance individuals hold than criminal involvement itself. In this sense, the appendix results reinforce the broader interpretation of the paper: insurance outcomes in this sample appear to be shaped more by labor market attachment and persistent disadvantage than by short run changes in offending.

These findings should be interpreted considering several limitations. First, the analysis is based on a specific high risk population of serious justice involved youth from two U.S. cities, so the results should not be generalized to the broader youth population. Second, criminal involvement is measured using self reported offending, which captures a broader range of behavior than administrative records but may also be affected by recall error or reporting bias. Third, while fixed effects address time invariant unobserved heterogeneity, they do not eliminate all potential endogeneity arising from time varying omitted factors. For these reasons, the results are best interpreted as evidence on how the crime insurance relationship changes once richer controls and within individual comparisons are used, rather than as definitive proof of a causal effect.

These findings complement the existing literature in important ways. While studies such as Markowitz et al. (1991) and Grossman (1972) establish that labor market attachment and socioeconomic resources are the primary determinants of insurance coverage, this paper extends that framework to a justice involved population and shows that criminal involvement adds little explanatory power once individual heterogeneity is accounted for. The results also speak to the broader literature on criminal records and labor market outcomes. Agan et al. (2025) and Gilleskie et al. (2025) both document how criminal records generate persistent disadvantages through employment and institutional channels and the findings here suggest those same channels, rather than offending itself, are what ultimately shape insurance access.

Overall, the paper contributes to the literature by examining a reverse link that has received much less attention: not how health or insurance affects crime, but how criminal involvement is

associated with insurance coverage over time. The main empirical contribution is to show that, in this sample, the negative cross-sectional relationship between crime and insurance largely reflects underlying differences across individuals rather than meaningful within person changes in offending. More broadly, the results suggest that policies aimed at strengthening labor market attachment and reducing persistent socioeconomic disadvantage may be more relevant for improving insurance access among justice involved youth than changes in criminal involvement alone.

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9 Appendices:

Appendix A1: Insurance Type Outcomes (Fixed Effects Full Model)

Variable	Parental insurance	Private insurance	Public insurance
Criminal involvement (Any crime)	0.015	-0.005	0.001
Age	-0.288***	-0.047	-0.209
Age squared	0.007***	-0.000	0.003
Health status: Good	-0.009	0.007	0.003
Health status: Fair	-0.012	0.018	-0.000
Health status: Poor	0.001	-0.093	0.041
Weeks worked legally	-0.000	0.003***	-0.002***
Future orientation	0.010	-0.001	-0.013
Perceived opportunity / success	-0.004	-0.020	0.024
Chance of staying out of trouble	0.005	0.005	-0.003
Low chance of success	0.001	-0.022**	-0.005
School aspiration	-0.007	0.003	0.008
Wave 9	-0.070***	0.052	0.094**
Wave 10	-0.111**	0.092	0.187**
Wave 11	-0.175**	0.137	0.302**
Constant	2.961***	1.263	3.269*
Observations	4394.000	4394.000	4394.000
Within R-squared	0.058	0.030	0.014

Note: The dependent variables are indicators for parental insurance, private insurance, and public insurance. All specifications report individual fixed-effects estimates using the full set of controls. Criminal involvement is measured as a binary indicator equal to one if the respondent reports any criminal activity in the previous 12 months. All models include age, age squared, health status, weeks worked legally, behavioral expectations, and wave fixed effects. Standard errors are clustered at the individual level. * p<0.10, ** p<0.05, *** p<0.01.